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Chemical physicists quantitatively model electron interactions in real quantum materials

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what other electrons are doing at any moment that they cannot be averaged together.

The team describes the new technique and results in a [paper](#) published in *Science*. The lead authors are Linqing Peng (Ph.D.) and Tianyu Zhu of Yale University. Both Peng and Zhu started working on the project in the lab of Garnet Chan, Bren Professor of Chemistry and director of the Rudolph A. Marcus Center for Theoretical Chemistry at Caltech.

"It is now possible to predict the properties of some complicated materials purely through computation without referring to experiment," says Chan, who is the senior author of the paper and a Simons Investigator in Physics. "These first materials that we have studied are like a baby step, or a prototype problem, along the way to more complex phenomena such as high-temperature superconductors and quantum magnets."

The Kondo effect—a many-body problem

For many of the materials used in computing, including semiconductors such as silicon, interactions between electrons are so weak that they are insignificant to understanding overall behavior. But to learn more about so-called strongly correlated materials needed for quantum applications, pinning down how electrons interact and bounce off each other is crucial.

The Kondo effect comes into play in the simplest example of such a strongly correlated material: the case where a single magnetic atom, such as iron or manganese, is embedded as an impurity in a metal, such as copper.

When that bulk material is cooled below a certain temperature (known as the Kondo temperature), something odd happens. When a normal metal is cooled, its electrical resistance decreases steadily, making the material a better conductor at lower and lower temperatures. When a metal with a magnetic impurity reaches its Kondo temperature, its resistance reaches a minimum and then increases.



fluctuates freely. But as the temperature drops, electrons in the metal begin to interact strongly with the atom's spin and can flip their own spin to cancel out a bit of the atom's magnetic moment.

This extra scattering interaction causes the characteristic leveling off and then an increase in resistance associated with the Kondo effect. Eventually, when the whole cloud of electrons gets involved, it masks the atom's magnetism completely.

The electrons in the metal enveloping the atom collectively arrange themselves into a cloud that "screens," or cancels out, the atom's magnetism.

Solving the problem quantitatively

The Kondo effect is one of the most intensively studied problems in quantum many-body physics, in part because it is relatively simple to state. It is used as something of a benchmark that researchers must show they can approximate for new theoretical and computational methods to gain acceptance. But pinning down the precise way in which resistance decreases and then comes back up, or the exact temperature at which the change occurs for a given impurity in a real material, has not been possible until now.

Previous methods have simplified the problem by reducing a material's electronic structure to a small number of orbitals—spaces around an atom's nucleus where electrons are most likely to be found—and then applying an approximate mathematical model to that reduced system.

Chan and his team took a different approach. They used technology that had been developed to accurately describe molecules in the quantum chemistry field and applied it to this material setting. This allowed them to fully describe the magnetic atom impurities as though they were molecules without having to simplify their interactions.

In the paper, the researchers provide results for seven different transition-metal atoms embedded in copper. For most elements, the new method's predictions exceeded the accuracy of model-

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Key concepts

Electronic structure

Magnetism

Transport phenomena

Quantum many-body systems

Strongly correlated systems

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